

Project Proposal: AI-Assisted Clinical Evaluation System for Maritime Nursing

1. Executive Summary

We are going to develop an intelligent, offline application designed to assist nurses in simulation training. Currently, the evaluation of Radio Medical Records (RMR) relies on pre-fabricated responses that lack specificity and clinical depth. Our solution replaces this with an AI-agent-driven system that reviews clinical documentation for accuracy, identifies missing data according to Danish Maritime Authority standards, and generates adaptive feedback to improve nursing triage and professional reporting.

2. Core Technology Stack

To ensure the system remains functional in offline maritime environments while providing high-performance analysis, we are going to use the following stack:

- **Frontend: React** will be used to build a professional, multi-page interface. This UI will strictly follow the official RMR structure, including sections for A: Airway, B: Breathing, C: Circulation, D: Disability, E: Expose, and the Observation Chart.
- **Backend: Python** with **FastAPI** will serve as the engine for our clinical logic. Python allows us to integrate advanced AI libraries and manage complex medical data validation.
- **Vector Database: ChromaDB** will act as our "Clinical Knowledge Base." We are going to store reference values such as normal blood pressure (120-140/60-90), pulse (60-80), and oxygen saturation (95-100%).
- **Primary Database: MongoDB** will be utilized to store simulation logs, allowing nurses and instructors to track performance history and longitudinal progress.

3. AI Agent Architecture & Reasoning

We are going to move beyond simple automation by creating a **Reasoning Agent** in Python. When a nurse submits a report, the agent will perform the following steps:

- **Clinical Validation:** The agent will scan the report to ensure all critical assessments were performed. For instance, if a nurse marks a patient as having "insufficient gasping," the agent checks if CPR was initiated.
- **Contextual Cross-Referencing:** Using RAG (Retrieval-Augmented Generation) through ChromaDB, the agent compares the recorded vitals against the nurse's

"General Condition" score. If a nurse scores a patient as "1 - Unaffected" while the heart rate is outside the 60-80 bpm range, the agent flags this inconsistency.

- **Adaptive Feedback Generation:** Using a local LLM (via Ollama), the agent will generate a situation-specific response. Instead of a template, it will provide a professional critique: *"The recorded pupil reaction was abnormal, but no description was provided in the findings section as required. Ensure neurological observations are fully documented."*

4. Quality Assurance with Promptfoo

To ensure our AI agent remains a reliable "Medical Auditor," we are going to implement **Promptfoo** for automated testing:

- **Adversarial Testing:** We will create test cases with intentionally conflicting data (e.g., normal blood pressure but "Pale" skin color) to see if the agent detects the discrepancy.
- **Benchmark Validation:** We are going to use Promptfoo to ensure the agent's advice consistently aligns with the Danish Maritime Authority's guidelines for oxygen administration (e.g., using a Hudson mask for >10 l/min).
- **Output Consistency:** Promptfoo will help us verify that the feedback generated is always professional, medically sound, and free of "hallucinations" or errors.

5. Proposed Development Roadmap

Phase	Task	Technology
Setup	Configure local LLM and Vector Store with RMR protocols.	Python, ChromaDB
Frontend	Build the 3-page RMR digital form for nurse data entry.	React
Intelligence	Develop the reasoning agent to detect clinical errors.	FastAPI, Ollama
Testing	Run automated clinical scenarios via Promptfoo to verify accuracy.	Promptfoo
Deployment	Finalize the offline package for simulation use.	Docker / Local Host